is shown in the figure 6.5. From the graph it is clear that the volume of the gas linearly increases with temperature at a given pressure. Such lines are called isobars. It can be expressed by the following straight line equation.

V = mT + C where T is the temperature in degree Celsius and m & C are constants.

When T= 0 °C the volume becomes V_0. Hence, V_0 = C and slope of the straight line m is equal to ΔV / ΔT. Therefore the above equation can be written in the following form.

$$V = \left(\frac{\Delta V}{\Delta T}\right)T + V_0 \quad \text{-------- (6.7)}$$

(n, P are constant)

Divide the equation 6.7 by V_o

$$\frac{V}{V_0} = \frac{1}{V_0}\left(\frac{\Delta V}{\Delta T}\right)T + 1 \quad \text{-------- (6.8)}$$

Charles and Gay Lussac found that under constant pressure, the relative increase in volume per degree increase in temperature is same for all gases. The relative increase in volume per ° C (α)is equal to $\frac{1}{V_0}\left(\frac{\Delta V}{\Delta T}\right)$.

Therefore

$$\frac{V}{V_0} = \alpha T + 1$$

$$V = V_0(\alpha T + 1) \text{ ------ (6.9)}$$

Charles found that the coefficient of expansion is approximately equal to 1/273. It means that at constant pressure for a given mass, for each degree rise in temperature, all gases expand by 1/273 of their volume at 0 °C

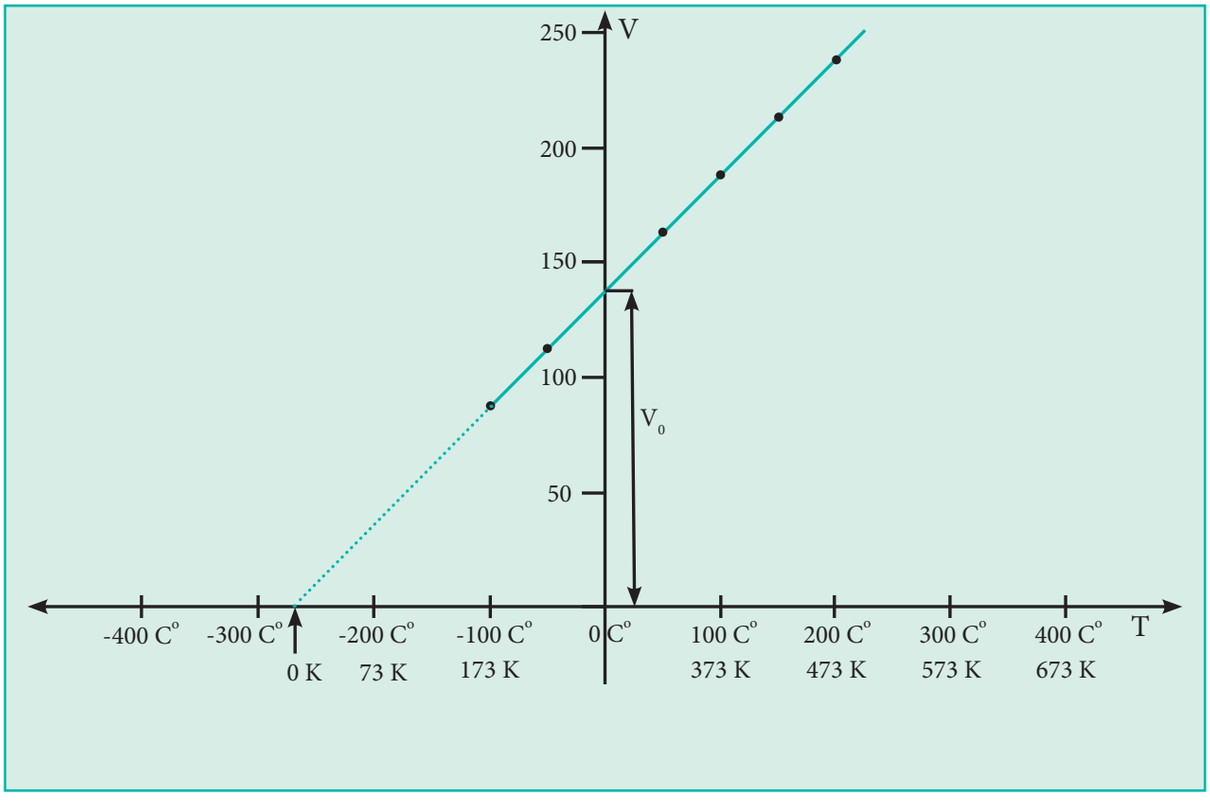


Figure. 6.5 Plot of volume vs temperature for an ideal gas

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